

Debasish Chakroborti (Joy), Ph.D.

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Career Objectives

- My research focuses on **AI assisted software maintenance**, with the goal of delivering **issue free software on the versions people actually use**. I pursue and continue work in legacy system monitoring, technical debt management, continuous maintenance, cybersecurity maintenance, resource optimization, and software sustainability, in close collaboration with industry. I also train the next generation of computing professionals through teaching, and I aim to expand that impact further.

Education

- 2020/24 ■ **Doctor of Philosophy, Computer Science/Software Engineering**, University of Saskatchewan, Canada. Thesis titled *Automated Backporting for Efficient Version Management in Software Repositories*. Advisors Prof. Dr. Chanchal K. Roy and Prof. Dr. Kevin Schneider. January 2020 to December 2024. Course mark 89%. Awards include Carl McCrosky Innovation Scholarship for Graduate Studies (2023), GSA Mark Kroeker Exceptional Student Leadership Award (2023), Ph.D. Citizenship Award (2024), Education Graduate Award from Friends Club of Saskatoon (2024) for remarkable contribution in research and innovation, Faculty/Departmental Scholarship, SOAR, SEMLA, and CSGC Best Poster Awards, Ivan and Margaret Toutloff Award, and Graduate Travel Awards (Seattle, Melbourne, Montreal).
- 2017/19 ■ **Master of Science, Computer Science/Big Data**, University of Saskatchewan, Canada. Thesis titled *An Intermediate Data-driven Methodology for Scientific Workflow Management System to Support Reusability*. Advisors Prof. Dr. Chanchal K. Roy and Prof. Dr. Kevin Schneider. September 2017 to November 2019 (thesis defended 12 November 2019, requirements completed 23 November 2019). Course mark 89%. Awards Saskatchewan Innovation and Opportunity Scholarship and University of Saskatchewan Faculty Stipend.
- 2008/12 ■ **Bachelor of Science, Computer Science and Engineering**, Khulna University, Bangladesh. Thesis titled *Web Service Performance Enhancement for Portable Devices Modifying SOAP Security Principle*. Supervisor Dr. Masud Rahman. January 2008 to May 2012. CGPA 3.73/4.00 (1st class 3rd). **Class Representative** (Term 3.1), CSE Discipline. Awards Innovation Award, Khulna University, Dean's Merit Scholarship / Merit List Scholarships (2008 to 2012), 3rd place at Digital Innovation Fair, Khulna (2010) for Geographic Information System Website and Khulna City Corporation Holding Tax Management System, and contestant of Citi Financial IT Case Competition (organized by D.Net, 2010). Coursework projects include Simple Arithmetic Evaluation (Java), Mobile Photo Editor (J2ME), Geographic Information System Website, Khulna City Corporation Holding Tax Management System, KU Alumni Website, E-library Website, and Reunion website (PHP).
- 2006 ■ **Higher Secondary Certificate (H.S.C.), Science**, Police Line School and College, Rangpur, Bangladesh (Rajshahi Board).
- 2004 ■ **Secondary School Certificate (S.S.C.), Science**, Gayabari School and College, Dimla, Nilphamari, Bangladesh (Rajshahi Board). Recognized among the **top 10 students in Dimla Upazila** for the S.S.C. result. **Winner**, Dimla Upazila general knowledge competition (Grade 9 / Class IX), as part of a **4-person team**.

Employment History and Projects

- 2025/26 ■ **Consultant (AI Automation Team), Cloud Linux Software, Inc., USA**. Research advisory on AI based automated backporting tooling.
- 2024/Pres. ■ **Instructor, School of Computing and Digital Innovation, Saskatchewan Polytechnic** (Regina Campus). Computer Systems Technology and Cloud Computing & Blockchain. Selected faculty participant, China Henan (ZJTIE) Transnational Education Project (2026): teaching and conducting labs in person in China, faculty meetings for collaboration, cultural exchange, and preparation for faculty travel abroad (institutional travel training and partnership coordination).
- 2024/Pres. ■ **Living Skies Postdoctoral Fellow, Department of Computer Science, University of Saskatchewan**. NSERC CREATE Software Analytics Research (SOAR) project. Automated software backporting and AI driven development tools.
- 2025 ■ **Sessional Lecturer, Department of Computer Science, University of Saskatchewan**. Taught **CMPT 145:3** Principles of Computer Science (Spring/Summer 2025, MT1 and MT2).
- 2024 ■ **Software Engineer (Intern), Push Interactions**. Enhanced <https://www.wilger.net>.
- 2023 ■ **PhD Intern (Remote), Synesis IT GmbH, Germany**. API planning and big data databases for Traffic Management Intelligent Transportation Systems.
- 2022/24 ■ **GSA Office Web Coordinator (Part-time), Graduate Students' Association, UofS**. Maintained the GSA website at <https://gsa.usask.ca> and added pages for services and events. HTML, JavaScript.
- 2022/24 ■ **Software Engineer (Part-time), Saskatchewan Education Leadership Unit**. Maintained the Placement Application at <https://placement.usask.ca> and developed a matching model for Teacher Candidates and Mentors.

Employment History and Projects (continued)

- 2017/24 ■ **TA / RA, M.Sc. and Ph.D., University of Saskatchewan.** Teaching Assistant / Student Assistantship (PSAC 40004) for CMPT 214 (Fall 2023), 145 (Summer 2023), 370 (Fall 2022), 280 (2022 to 2023), 270 (2020 to 2022), 141, and Database Management Systems. Plant Phenotyping and Imaging Research Centre and Global Institute for Food Security. Developed **RISP** and **RISPs** data reusability algorithms (Hadoop/Spark). Built SE and imaging tools **Baczer**, **ReBack** (CNN), **BackTrans** (LLM), **BackSlice**, and **InsCount** (UNet).
- 2017/24 ■ **Lecturer (on study leave), CSE Discipline, Khulna University, Bangladesh.** On study leave for graduate studies at the University of Saskatchewan.
- 2013/17 ■ **Lecturer, CSE Discipline, Khulna University, Bangladesh.** Teaching and research. Implemented websites for Khulna University (<https://ku.ac.bd>), ICCIT (<https://iccit.org.bd/2013/>), and the Convocation 2016 Registration Website.
- 2014/16 ■ **Part-time Lecturer, North Western University, Khulna, Bangladesh.** Computer Science courses for the BA program.
- 2014/16 ■ **Instructor (Part-time), Higher Secondary Teachers' Training Institute (HSTTI), Khulna, Bangladesh.** ICT courses for high school teachers.
- 2013 ■ **Software Engineer, Nascenia Limited, Bangladesh** (29 May to 30 September 2013). **SnapKnot** wedding photographer directory (<https://snapknot.com/>). Ruby on Rails, ASP.NET, C#, Git.
- 2012/13 ■ **Software Engineer, Divine IT Limited, Bangladesh.** ERP systems (<https://www.divineit.net>), including Lines Pay, Sphere, Prism, and Centroid. Django, Mysql, Postgresql, sqlite, SVN.

Research Publications

Journal Articles

- 1 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2024c). Code propagation in stable version management: A systematic literature review. (*Under review*). Manuscript under review.
- 2 **Chakroborti, D.**, Schneider, K. A., & Roy, C. K. (2024). ReBack: Recommending backports in social coding environments. *Automated Software Engineering*, 31(1), 18. doi:10.1007/s10515-024-00416-1
- 3 **Chakroborti, D.**, Nath, S. S., Schneider, K. A., & Roy, C. K. (2022). Release conventions of open-source software: An exploratory study. *Journal of Software: Evolution and Process*, e2499. JSME-22. doi:10.1002/smr.2499
- 4 **Chakroborti, D.**, Roy, B., & Nath, S. S. (2021). Designing for recommending intermediate states in a scientific workflow management system. *Proceedings of the ACM on Human-Computer Interaction*, 5(EICS). doi:10.1145/3457145
- 5 Herbold, S., Trautsch, A., Ledel, B., Aghamohammadi, A., Ghaleb, T., Chahal, K., **Chakroborti, D.** et al. (2021). A fine-grained data set and analysis of tangling in bug fixing commits. *Empirical Software Engineering*. doi:10.1007/s10664-021-10064-8
- 6 Bairagi, A. K., Mondal, S., & **Chakroborti, D.** (2017). Securing bangla text communication using image steganography with dynamic substitution in IoT environment. *Khulna University Studies*, 14. KUS: 16/30:010816.

Conference Proceedings

- 1 **Chakroborti, D.** (2026b). Backporting battles on the branches: Agentic vs human fixes. In *International conference on software engineering (ICSE)*. Submitted.
- 2 Tasnim, J., **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2025). An insight into the technical debt–fix trade-off in software backporting. In *Proceedings of the 35th IEEE international conference on collaborative advances in software and computing (CASCON 2025)*, York University, Toronto, ON, Canada: IEEE.
- 3 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2024a). A study of backporting code in open-source software for characterizing changesets. In *2024 IEEE/ACM 46th international conference on software engineering: Companion proceedings (ICSE-companion '24)*. doi:10.1145/3639478.3643079
- 4 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2024b). BackSlice: Achieving accurate and essential propagation through backport slicing. In (*submitted*). Submitted.
- 5 Tasnim, J., **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2023). How does quality deviate in stable releases by backporting? In *45th international conference on software engineering (ICSE 2023)*. May 14–20, Melbourne, Australia.
- 6 **Chakroborti, D.**, Schneider, K. A., & Roy, C. K. (2022). Backports: Change types, challenges and strategies. In *30th IEEE/ACM international conference on program comprehension (ICPC 2022)*. May 16–17. doi:10.1145/3524610.3527920
- 7 Bhattacharjee, A., Nath, S. S., Zhou, S., **Chakroborti, D.**, Roy, B., Roy, C. K., & Schneider, K. (2020). An exploratory study to find motives behind cross-platform forks from software heritage dataset. In *Proceedings of the 17th international conference on mining software repositories* (pp. 11–15). doi:10.1145/3379597.3387512
- 8 **Chakroborti, D.**, Mondal, M., Roy, B., Roy, C. K., & Schneider, K. A. (2018). Optimized storing of workflow outputs through mining association rules. In *2018 IEEE international conference on big data (big data)* (pp. 508–515). doi:10.1109/BigData.2018.8622351

- 9 **Chakroborti, D., & Nath, S. S.** (2017). Web service performance enhancement for portable devices modifying SOAP security principle. In *2017 20th international conference of computer and information technology (ICCIT)* (pp. 1–7).
 doi:10.1109/ICCITECHN.2017.8281799
- 10 Bairagi, A. K., & **Chakroborti, D.** (2015). Trust based D2D communications for accessing services in internet of things. In *2015 18th international conference on computer and information technology (ICCIT)* (pp. 50–54).
 doi:10.1109/ICCITechN.2015.7488041

Under Submission / In Preparation

- 1 **Chakroborti, D.** (2026a). An insight into backporting changes: An empirical study with pull-based development. In *(to be submitted)*. To be submitted.
- 2 **Chakroborti, D.** (2026c). Beyond correctness: Technical debt and carbon footprint of LLM-agent versus human backports. In *(to be submitted)*. To be submitted.
- 3 **Chakroborti, D.** (2026d). BranchBench and BranchBridge: Patch-level evaluation for CVE backport adaptation. In *(to be submitted)*. To be submitted.
- 4 Tasnim, J., **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2026). Impact of backporting on the quality of stable software releases: A comparative analysis. In *(to be submitted)*. To be submitted.

Books and Chapters

- 1 **Chakroborti, D.**, Roy, B., Mondal, A., Mostaeen, G., Roy, C. K., Schneider, K. A., & Deters, R. (2020). A data management scheme for micro-level modular computation-intensive programs in big data platforms. In R. Alhaji, M. Moshirpour, & B. Far (Eds.), *Data management and analysis: Case studies in education, healthcare and beyond* (pp. 135–153).
 doi:10.1007/978-3-030-32587-9_9

Posters

- 1 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2022). Backporting for version management: Automated changesets integration in software repositories. In *Symposium on software engineering for machine learning applications (SEMLA)*. Best Poster (3rd).
- 2 Tasnim, J., **Chakroborti, D.**, Roy, C. K., Schneider, K. A., & Mondal, M. (2022). Detecting and visualizing quality aspects in backporting pull requests. In *Symposium on innovations in computer science and applied computing (ICSAC)*. Best Poster (2nd).
- 3 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2021). Identifying and integrating changesets in backporting. In *The annual software analytics research (SOAR) symposium*. Best Poster.

Talks, Posters & Conference Visits

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| 2025 |  CASCON 2025 (York University, Toronto, Canada). Conference talk on the technical debt fix trade off in software backporting (IEEE CASCON 2025 paper with Tasnim, Roy, and Schneider). Supported by Saskatchewan Polytechnic Professional Development funding. |
| 2023 |  ICSE 2023 (Melbourne, Australia). Poster presentation <i>How Quality Deviates in Stable Releases through Backporting</i> (ICSE 2023 / NIER related work with Tasnim, Roy, and Schneider). Attended the main conference and co located events (14 to 20 May 2023). Graduate travel award support. |
| 2022a |  SEMLA 2022 (Polytechnique Montréal, Montréal, Canada). Poster presentation <i>Backporting for Version Management</i> (Best Poster, 3rd). Graduate travel award support. |
| 2022b |  ICPC 2022 (co located with ICSE 2022). Conference talk <i>Backports Change Types, Challenges and Strategies</i> (research paper with Schneider and Roy). Presented in the scheduled ICPC program session. |
| 2022c |  GSA Research Conference 2022 (University of Saskatchewan). Poster presentation. |
| 2022d |  Research Fest 2022 (CSGC, University of Saskatchewan). Poster presentation. |
| 2021 |  SOAR Symposium 2021 (University of Saskatchewan). Poster presentation <i>Identifying and Integrating Changesets in Backporting</i> (Best Poster). |
| Local |  CSGC Research Fest / ICSAC . Distinguished speaker recognition for a Research Fest talk. Also Lead Organizer for the 7th ICSAC / Research Fest. |
| 2018/19 |  P2IRC poster presentations (Plant Phenotyping and Imaging Research Centre, University of Saskatchewan / GIFS), 2018 and 2019. |
| 2018 |  IEEE Big Data 2018 (Seattle, WA, USA). Conference talk <i>Optimized Storing of Workflow Outputs through Mining Association Rules</i> (primary author). Presentation, 10 to 13 December 2018. Graduate travel bursary support. |
| 2017 |  ICCIT 2017 (University of Asia Pacific, Dhaka, Bangladesh). Conference talk <i>Web Service Performance Enhancement for Portable Devices Modifying SOAP Security Principle</i> (with Nath). 20th International Conference on Computer and Information Technology, 22 to 24 December 2017. |

Talks, Posters & Conference Visits (continued)

- 2015 ■ **ICCIT 2015** (Military Institute of Science and Technology, Dhaka, Bangladesh). Conference talk *Trust Based D2D Communications for Accessing Services in Internet of Things* (with Bairagi). 18th International Conference on Computer and Information Technology, 21 to 23 December 2015.

Research Grants & Funding

- Ongoing ■ **NSERC Alliance Advantage Grant**. Postdoctoral Fellow. Contributing research as a Living Skies Postdoctoral Fellow on an NSERC Alliance Advantage project (industry academia collaboration).
- 2025/26 ■ **Cloud Linux Software Consultancy Fund**. AI Automation Consultant. Industry consulting on AI based automated backporting tooling.
- 2025 ■ **Saskatchewan Polytechnic Professional Development Fund**. Faculty PD support to present research at a conference in Toronto.
- Support ■ **Funding application support**. Proposal contributor. Assisted M.Sc. mentor with drafting a portion of a research funding application.

Media Mentions & Publicity

- 2023 ■ **USask Computer Science news**. Featured among Graduate Student Award Recipients (Carl McCrosky Innovation Scholarship). <https://www.cs.usask.ca/news/2023/graduate-student-award-recipients.php>
- Lab ■ **Software Research Lab (SRLab) awards**. Listed for Ph.D. Citizenship Award (2024), McCrosky Scholarship (2023), GSA Mark Kroeker Exceptional Student Leadership Award (2023), and Saskatchewan Innovation and Opportunity Scholarship (Ph.D., 2019). <https://srlab.usask.ca/awards/>
- CSGC ■ **Computer Science Graduate Council**. Listed as CSGC President (and earlier VP Social) on previous committees. <https://csgc.usask.ca/previous-committees/>
- LinkedIn ■ **LinkedIn features / announcements**. McCrosky Scholarship announcement. Ph.D. Citizenship Award share of CS news. CSGC/ICSAC Research Fest leadership. CASCON paper acceptance. ICSE Companion paper share. Profile at <https://www.linkedin.com/in/joydeba/>
- Thesis ■ **USask Spring Convocation 2025**. Listed for Doctor of Philosophy (Computer Science). <https://students.usask.ca/documents/convocation/spring-2025.pdf>
- HARVEST ■ **USask HARVEST**. Ph.D. thesis record *Automated Backporting for Efficient Version Management in Software Repositories*. <https://harvest.usask.ca/items/59b8a934-f345-4b99-817d-b104024a5ac9>
- Social ■ **Additional channels**. Further announcements are shared via University of Saskatchewan / Department of Computer Science social accounts and personal LinkedIn, Facebook, and Instagram.

Professional Service

- PC ■ **PC Member**, Mining Software Repositories (MSR 2026). Data/Challenge track. **PC Member**, SOAR Symposium 2025 (University of Saskatchewan).
- Org ■ **Lead Organizer**, 7th Symposium on Innovations in Computer Science and Applied Computing (ICSAC / Research Fest). **Judge**, Research Fest 2025 (Computer Science Graduate Council, Department of Computer Science, USask). **Judge**, Images of Research contest, USask (2024). **Judge**, GSA Graduate Research Elevator Pitch Competition (2024). **Judge**, GSA Graduate Research Conference (2023).
- Journals ■ **Journal of Software Evolution and Process (JSME/SMR)**. Reviewer (2023 to 2025, reviewer certificate for 2023). **Journal of Systems and Software (JSS)**. Reviewer / sub reviewer (2020, 2022, 2025).
- 2026 ■ **ASE 2026**, **FSE 2026**, **ICPC 2026**, and **ICSE 2026 SEIS**. Reviewer / sub reviewer.
- 2025 ■ **CHI 2025** and **SANER 2025**. Reviewer / sub reviewer.
- 2024 ■ **ASE 2024** and **CASCON 2024**. Sub reviewer.
- 2023 ■ **ICSME 2023** and **SANER 2023**. Sub reviewer.
- 2022 ■ **ISEC 2022**. Sub reviewer (multiple papers).
- 2020 ■ **ASE 2020**, **ICSE 2020** (Tool Demo), **IWSC 2020**, and **ICSA 2020**. Sub reviewer.
- Internal ■ **Internal manuscript reviews**. Detailed reviews for lab colleagues' drafts (conference/journal submissions and thesis chapters).

Collaborations

- SRLab ■ **Software Research Lab (SRLab)**, University of Saskatchewan. Ongoing research collaboration on AI assisted software maintenance, automated backporting, and software analytics (Ph.D. and Living Skies Postdoctoral Fellowship).
- iSE ■ **Interactive Software Engineering Lab (iSE Lab)**, University of Saskatchewan. Collaboration on scientific workflow / software analytics research and related lab activities.
- Industry ■ **Cloud Linux Software, Inc.** Industry collaboration on AI based automated backporting tooling (AI Automation Consultant). **Push Interactions**. Industry collaboration on web application enhancement (Wilger).

Collaborations (continued)

- P2IRC ■ **Plant Phenotyping and Imaging Research Centre (P2IRC)**, University of Saskatchewan / GIFS. Collaboration on scientific data management and imaging tooling (e.g., RISP/RISPs, InsCount).
- Water ■ **Water Security project** (USask CFREF / water security research collaboration). Collaboration supporting scientific computing and software analytics needs for water security research.

Student Supervision & Mentoring

- M.Sc. ■ **M.Sc. student mentoring**. Mentored **Jarin Tasnim** (University of Saskatchewan) on research related to software backporting and quality analysis (co authored publications).
- UG ■ **Undergraduate summer students**. Supervised 3 undergraduate summer research students.
- TAs ■ **Teaching assistants**. Guided 3 TAs in support of sessional lecturer course **CMPT 145**, Department of Computer Science, University of Saskatchewan (Spring/Summer 2025).

Teaching & Curriculum

- 2024/... ■ **Instructor**, School of Computing and Digital Innovation, Saskatchewan Polytechnic (Regina Campus). **Computer Systems Technology** courses COOS 190 (Server Administration), COOS 294 (Cloud Infrastructure Administration), COOS 295 (Systems Administration 2), COET 295 (Emerging Technologies), CNET 184, CWEB 280, and CMPT 145. **Cloud Computing & Blockchain** courses CCMP 600 to 606, DEVP 600, and PROJ 611.
- Curriculum ■ **Curriculum and lab design** (Saskatchewan Polytechnic). Designed and/or rewrote course and lab materials for most Sask. Poly courses taught (above).
- LIFT ■ **LIFT Certificate Program**, Saskatchewan Polytechnic. Statement of Achievement for Learners, Instruction & Fundamentals of Teaching (**90 hours**, May 2026). Instructional ePortfolio (LIFT 1008) at https://online.saskpolytech.ca/d2l/eP/presentations/presentation_preview_popup.d2l?presId=63930&pageId=3744&ou=6606. Local backup at <https://joydeba.github.io/teaching-materials/lift-eportfolio/>. Statement PDF at <https://joydeba.github.io/teaching-materials/lift-statement-of-achievement.pdf>.
- Dossier ■ **Teaching Dossier**. Philosophy, course portfolio, effectiveness evidence, sample materials, innovation, and continuous improvement. <https://joydeba.github.io/teaching-materials/teaching-dossier.pdf>
- ZJTIE ■ **Selected Faculty Participant**, China Henan (ZJTIE) Transnational Education Project, Saskatchewan Polytechnic (2026). Teaching and conducting labs in person in China, faculty meetings for collaboration, cultural exchange, and preparation for faculty travel abroad (institutional travel training and partnership coordination).
- NIO ■ **New Instructor Orientation (NIO)**, Saskatchewan Polytechnic. Statement of Achievement, 18 hours (14 to 16 Jan 2025). <https://joydeba.github.io/teaching-materials/nio-statement-of-achievement.pdf>
- CETL ■ **1-day Training on Pedagogy** (Induction for the Beginners), Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP). 12 September 2014.
- CETL1 ■ **3-day Training on Pedagogy** (Module 1 Teaching & Learning), Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP). 11 to 13 June 2015. Certificate at <https://joydeba.github.io/teaching-materials/ku-cetl-pedagogy-2015.jpg>.
- CETL2 ■ **2-day Training on Pedagogy** (Module II Teaching & Learning), Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP). 30 to 31 December 2016.
- GTI ■ **14th Foundation Training for University Teachers (FTUT)**, Graduate Training Institute (GTI), Bangladesh Agricultural University, Mymensingh (UGC sponsored). 150 hours, 24 May to 22 June 2014. Nominated as Lecturer, CSE, Khulna University. Certificate at <https://joydeba.github.io/teaching-materials/gti-ftut-14th-certificate.jpg>. Course modules at <https://joydeba.github.io/teaching-materials/gti-ftut-14th-course-modules.jpg>.
- 2025 ■ **Sessional Lecturer**, Department of Computer Science, University of Saskatchewan. **CMPT 145.3** Principles of Computer Science (Spring/Summer 2025, sections MT1 and MT2, CRNs 41442 and 60860). Guided 3 TAs.
- SLEQ ■ **Student Learning Experience Questionnaire (SLEQ)**, USask. End of course report for **CMPT 145.01** (CRN 41442, Spr/Sum 2025). Audience 38. Responses 3 (7.9%). **Below Response Threshold** (closed ended scores not released, qualitative comments only). Report at <https://joydeba.github.io/teaching-materials/sleq-cmpt145-2025.pdf>.
- 2017/24 ■ **Teaching Assistant / Student Assistantship**, Department of Computer Science, University of Saskatchewan (PSAC Local 40004). **CMPT 214** (Fall 2023). **CMPT 145** (Summer 2023). **CMPT 370** (Fall 2022). **CMPT 280** (2022, 2023). **CMPT 270** (Fall 2020, Spring 2021, Fall 2021, Spring 2022). **CMPT 141**. Database Management Systems.
- 2014/16 ■ **Part-time Lecturer**, North Western University, Khulna, Bangladesh. Computer Science courses for the BA program.
- 2014/16 ■ **Instructor (Part-time)**, Higher Secondary Teachers' Training Institute (HSTTI), Khulna, Bangladesh. ICT courses for high school teachers.
- 2013/17 ■ **Lecturer**, CSE Discipline, Khulna University, Bangladesh. Courses included Structured Programming, Software Development Project, Advanced Programming Laboratory, and Software Engineering and Information System (lectures, labs, assignments, and exams).

Certifications & Professional Development

AWS	■ AWS Academy Educator . AWS Skill Builder digital badges / micro credentials (2024 to 2025), including AWS Cloud Practitioner Essentials, Designing Blockchain Solutions using Amazon Managed Blockchain, Amazon S3 Business Continuity and Disaster Recovery, AWS Managed Services (AMS) Advanced Disaster Recovery, Getting Started with Amazon ECS, Getting Started with DevOps on AWS, Amazon EKS Primer, Amazon EMR Getting Started, AWS Backup Primer, and AWS Billing and Cost Management.
Teach	■ LIFT (Learners, Instruction & Fundamentals of Teaching). Statement of Achievement, 90 hours (Saskatchewan Polytechnic, May 2026). New Instructor Orientation (NIO) . Statement of Achievement, 18 hours (14 to 16 Jan 2025). Reconciliation Education . Certificate of Completion, First Nations University of Canada (Indigenous awareness for the workplace). 4 Seasons of Reconciliation PD. Staff version (Oct 2025). Learning & Teaching Kickoff Day . Statement of Attendance (Aug 2024). WHMIS 2015 . Saskatchewan Polytechnic (issued Nov 2024, valid to Nov 2027).
CETL	■ 1-day Training on Pedagogy (Induction for the Beginners). Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP), 12 September 2014. 3-day Training on Pedagogy (Module I Teaching & Learning). Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP), 11 to 13 June 2015. 2-day Training on Pedagogy (Module II Teaching & Learning). Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP), 30 to 31 December 2016.
GTI	■ 14th Foundation Training for University Teachers (FTUT) . Graduate Training Institute (GTI), Bangladesh Agricultural University, Mymensingh (UGC sponsored). Residential foundation training for public university teachers, 150 hours , 24 May to 22 June 2014 (certificate no. 22440, nominated as Lecturer, CSE, Khulna University). Modules included Teaching Methods & Techniques (50 hours), Statistical Analysis, Research Methods and Project Cycle Management, Scientific Report Writing, Administration and Office Management, and Web Page Design and Hosting.
Betar	■ Industrial training , Bangladesh Betar, Khulna (5 to 26 January 2012). Radio broadcasting and control systems, magnetic tape to computerized conversion, power supply and central AC management, and internal network management.
Ethics	■ TCPS 2 CORE 2022 . Course on Research Ethics (Panel on Research Ethics, Certificate #0000795671, Feb 2022).
Policies	■ Workplace policies and compliance (acknowledged onboarding). CloudLinux Acceptable Use Policy (SOP_SEC_02), Employee Privacy Policy (SOP_DP_01), and Code of Ethics . Saskatchewan Polytechnic Code of Conduct (Policy 703) / institutional policy acknowledgment.
Coursera	■ Coursera . Machine Learning (Stanford University, Dec 2019). Deep Learning Specialization (deeplearning.ai, Jul 2020, five courses). Statistical Inference (Johns Hopkins University, course projects Sep 2020). Data Science Specialization coursework including Statistical Inference.
Members	■ ACM Student Member (membership no. 2631649, renewed through 31 Jan 2025). IEEE Member (joined Oct 2018).
HackerRank	■ HackerRank Role Certifications. Software Engineer (Mar 2024, ID 2179B8FECE54). Software Engineer Intern (Mar 2024, ID 668513C4226F).
Security	■ CloudLinux workplace security training suite (Jun 2025), covering cybersecurity essentials, anti phishing, social engineering, insider threats, data protection, remote work information security, and data breach response modules.

Skills

Languages	■ English and Bengali (strong reading, writing, and speaking).
Coding	■ Python, Java (J2SE, J2ME), C++, C#, Ruby, JavaScript, R, PHP, MATLAB, PROLOG, OpenGL, Assembly (8085, 8086 MASM), SQL, $\text{\LaTeX}$.
Frameworks	■ Django, Rails, Flask, ASP.NET, AngularJS, WordPress, Joomla, Spark, Hadoop, TensorFlow/PyTorch.
Databases	■ MySQL, PostgreSQL, SQL Server, CouchDB, SQLite.
Web	■ HTML, CSS, JavaScript, Ajax, XML, Apache Web Server, Git, GitHub, SVN.
Systems	■ Windows, Linux (Ubuntu), macOS. DNS and mail server configuration.
Focus	■ Software engineering, AI/ML for code, backporting, big data, teaching.

References

- Available upon request.